

1 简介【Introduction】

模组在使用过程中，受到比较严重的机械冲击或静电放电后，有极低的概率出现新增盲元，需要用户将新增盲元添加到模组固件的盲元表中。新增盲元的去盲元算法和以及保存功能已经在固件中实现，app 只需调用 IRCMD 库中的相关命令接口按照标定流程操作即可完成新增盲元的添加过程。

【During the use of the module, after a relatively severe mechanical shock or electrostatic discharge, there is a very low probability that new blind elements will appear, and the user needs to add the new blind elements to the blind element table of the module firmware. The algorithm for removing the blind elements and the saving function of the new blind elements have been implemented in the firmware. The app only needs to call the relevant command interface in the IRCMD library and follow the calibration process to complete the process of adding the new blind elements.】

2 自动盲元标定功能【Automatic dead pixel correction】

调用 IRCMD 中的 `dpcAutoCalibration` 函数可以进行自动盲元标定。标定时间越长，标定效果越好。

【Calling the `dpcAutoCalibration` function in IRCMD can perform automatic blind element calibration. The longer the calibration time, the better the calibration effect. 】

```
1  /**
2      * 自动盲元标定功能【**Automatic dead pixel correction**】
3      * auto calibrate dead pixel and save to flash in firmware.
4      *
5      * @param wait_time wait time of auto detect function (unit:s)
6      * @return see {@link CommonParams.IrcmdResult}
7      */
8  public int dpcAutoCalibration(int wait_time)
```

3 手动输入新增盲元坐标【Manually Enter the coordinates】

用户确认新增盲元坐标后，可通过下列步骤添加到模组固件的盲元表中：

【After confirming the coordinates of the dead pixels, you can perform the following steps to add them to the LUT of the module firmware: 】

3.1 Step1

调用函数 `addDeadPixelPoint` ,添加新的盲元到盲元表。

【Call function `addDeadPixelPoint` ,add a new blind element to the blind element table.】

```
1  /**
2      * 在盲元表中增加元素【Add dead pixel point temporarily】
3      * 需要发送{@link #saveSpiConfig(int)}的{@link
CommonParams#SPI_MOD_CFG_DEAD_PIX}才保存
```

```

4      * 【Need to send {@link #saveSpiConfig(int)} {@link
CommonParams#SPI_MOD_CFG_DEAD_PIX} before saving】
5      * <p>
6      * 其中，坐标x和坐标y分别代表图像的第x行和第y列（从1开始计数）
7      * 【Among them, the coordinate x and the coordinate y respectively
represent the xth row and yth column of the image (counting from 1)】
8      *
9      * @param x the pixel's x location 盲元位置【Blind location】
10     * @param y the pixel's y location 盲元位置【Blind location】
11     * @return 0 is success
12     * @see #removeDeadPixelPoint(int, int)
13     * @see #saveSpiConfig(int)
14     */
15     public int addDeadPixelPoint(int x, int y)

```

3.2 Step2

调用函数 `saveSpiConfig`，新的盲元表将被保存到flash。

【Call the function `saveSpiConfig` to save the new blind meta table to flash.】

```

1  /**
2      * 保存spi配置【save user config】
3      * 可以保存【Can save】
4      *
5      * @param cfgType 盲元类型【parameter type】
6      *      {@link CommonParams#SPI_MOD_CFG_ALL 将所有配置信息写入
flash【Write all configuration information to flash】}
7      *      {@link CommonParams#SPI_MOD_CFG_DEAD_PIX 将盲元信息写入
flash【Write blind meta information to flash】}
8      *      {@link CommonParams#SPI_MOD_CFG_PROPERTY_PAGE 将属性页信
息写入flash【Write property page information to flash】}
9      * @return 0 is success
10     * @see #addDeadPixelPoint(int, int)
11     * @see #removeDeadPixelPoint(int, int)
12     */
13     public int saveSpiConfig(int cfgType)

```

3.3 将盲元从盲元表中移除【Remove the blind element from the blind element table】

注：如果出现误操作，可以调用`removeDeadPixelPoint`将盲元从盲元表中移除。

【Note: If there is a misoperation, you can call `removeDeadPixelPoint` to remove the blind element from the blind element table.】

将盲元从盲元表中移除，调用 `removeDeadPixelPoint`。同理，也需要调用 `saveSpiConfig` 保存修改后的盲元表。

【To remove the blind element from the blind element table, call `removeDeadPixelPoint`. In the same way, you need to call `saveSpiConfig` to save the modified blind meta table.】

```

1  /**
2      * 在盲元表中去除元素【remove dead pixel point temporarily】

```

```

3      * 需要发送{@link #saveSpiConfig(int)}的{@link
CommonParams#SPI_MOD_CFG_DEAD_PIX}才保存
4      * 【Need to send {@link #saveSpiConfig(int)} {@link
CommonParams#SPI_MOD_CFG_DEAD_PIX} before saving】
5      * <p>
6      * 其中，坐标x和坐标y分别代表图像的第x行和第y列（从1开始计数）
7      * 【Among them, the coordinate x and the coordinate y respectively
represent the xth row and yth column of the image (counting from 1)】
8      *
9      * @param x the pixel's x location 盲元位置【Blind location】
10     * @param y the pixel's y location 盲元位置【Blind location】
11     * @return 0 is success
12     * @see #saveSpiConfig(int)
13     * @see #addDeadPixelPoint(int, int)
14     */
15     public int removeDeadPixelPoint(int x, int y)

```

4 计算盲元坐标参考方法【Reference method】

下文提供一种计算新增盲元的坐标的方法，仅供用户参考。

【The following provides a method to calculate the coordinates of newly added dead pixel, for reference only. 】

1. 将均匀平板放置在模组前，遮挡住全部视场；

【Place the uniform plate in front of the module to block all the field of view;】

2. 切换到TNR出图，采集1~30帧图像至一个文件夹内；

【Switch to TNR streaming and collect 1~30 frames of images into a folder;】

3. 脚本对图像进行尺寸3×3的中值滤波,原始像素值与滤波后像素值相差大于阈值的点标记为盲元;

【The script conducts median filtering with size of 3×3 for the image, and the points whose difference between the original pixel value and the filtered pixel value is greater than the threshold are marked as dead pixel. 】

注：若盲元数量与实际不符，可以更改盲元阈值。

【Note: If the number of dead pixel does not match the actual number, the threshold can be changed.】

5 matlab代码【matlab code】

```

1  %%dead pixel calculate
2  clear coordinate
3
4  T=40;%盲元阈值
5  Size=3;%计算尺寸
6
7  row=192;
8  col=256;
9  data2=zeros(row,col);
10 data1=zeros(row,col);
11

```

```

12 root=uigetdir('Ñ;õñîÄ%þ¼Ð');
13 x=[root, '/', '*.bin'];
14 getfilename=ls(x);
15 filename=cellstr(getfilename);
16 n=numel(filename);
17 if n>=60
18     n=60;
19 end
20 for i=1:n;
21     fileID=fopen([root, '/', filename{i}], 'rb+');
22     while ~feof(fileID)
23         [data]=fread(fileID, 'uint16');
24     end
25     fclose(fileID);
26     data=double(data);
27     for r=1:row
28         for c=1:col
29             data1(r,c)=data((r-1)*col+c);
30         end
31     end
32     data2=data2+data1;
33 end
34 data2=floor(data2./n);
35 figure;imshow(data2);caxis([min(min(data2));max(max(data2))]);title('Image')
36 ;
37 data2=padarray(data2, [1 1], 'replicate');
38 DP=int8(abs(data2-medfilt2(data2, [3 3]))>T);
39 figure;imshow(DP(2:193, 2:257));caxis([0 1]);title('DP');
40 n_DP=0;
41 for i=1:row
42     for j=1:col
43         if DP(1+i, 1+j)==1
44             n_DP=n_DP+1;
45             coordinate(n_DP, 1)=i;
46             coordinate(n_DP, 2)=j;
47         end
48     end
49 end
clearvars -except coordinate n_DP

```